Abbas Taqvi

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Professional Summary

- M.S. Data Science student at Milano-Bicocca with professional experience in data engineering and academic research in Bayesian inference. Built a Python Text2SQL system that reduced analyst turnaround by 41% and a financial data platform housing 30B+ records. Developed a PyTorch normalizing-flow framework for binary black-hole population modeling. Seeking a data science internship in Italy.

Education

- **Università degli Studi di Milano-Bicocca** — Milan, Italy
  Master of Science in Data Science — *Oct 2025 – Present*
- **Habib University** — Karachi, Pakistan
  Bachelor of Science in Computer Science; GPA: 3.31; 100% merit scholarship (HU-TOPS) — *Aug 2020 – May 2024*

Experience

- **Logiciel Services LLC** — United States (Remote)
  Data Engineer I — *May 2024 – Nov 2025*
  - **Text2SQL Analytics**: Built a Python multi-agent chatbot with LangChain, LangGraph and Trino, enabling business teams to query financial data in natural language through a REST API; reduced analyst turnaround by 41%.
  - **Financial Data Platform**: Built Spark (Scala) pipelines for 375+ daily FIX sessions, with Kafka, Airflow, S3 and AWS Glue. Simplified a 12-component architecture to 5; enabled sub-second queries over a 30B+ record lakehouse with appropriate filters.
  - **ETL Performance**: Developed a composable Rust framework for FIX-log processing, reducing daily runtime from 500 to 50 minutes (10× faster) and saving 400 EC2-minutes per run.
- **Pullflow Inc.** — San Jose, CA (Remote)
  DevOps Intern — *Jul 2023 – Aug 2023*
  - **Kubernetes Migration**: Migrated production services from AWS EKS to on-premise Kubernetes (MicroK8s): wrote Dockerfiles, Helm charts and a Traefik reverse-proxy config, with a zero-downtime cutover.
- **United Bank Limited** — Karachi, Pakistan
  Data Engineering Intern — *May 2023 – Jun 2023*
  - **Migration Pipeline**: Built an Informatica streaming pipeline into a Hive data lake and helped validate production AI pipelines against regional schemas across Pakistan, Qatar and Bahrain.
- **ImproData** — Pinedale, WY (Remote)
  Junior Data Engineer — *Jan 2021 – May 2022*
  - **Data Pipelines**: Built PySpark pipelines for cyber-security risk data on Azure Databricks using bronze, silver and gold layers. Automated JSON ingestion through GCS into BigQuery with GCP Cloud Functions, saving about 3 hours of daily manual work.

Projects

- **Astroflow: Bayesian Inference with Normalizing Flows**: Developed a PyTorch framework for binary black-hole population modeling, with neural summarizers and feature extractors. Reduced inference from roughly 300 minutes with traditional statistical methods to about 2 seconds; built a Django interface to organize experiments and visualize posterior samples.
- **Astroflow Research Poster**: Work featured as a poster at The Computing Conference 2025 (SAI Conferences), London: "Astroflow: Accelerating Parameter Inference of Binary Black Holes' Population using Flow-based Neural Networks."

Skills

- **Programming & Analysis**: Python, SQL, R, Scala, Rust; Pandas, NumPy, Matplotlib, Power BI
- **Machine Learning**: PyTorch, TensorFlow, JAX, NumPyro, normalizing flows, Bayesian inference
- **AI Applications**: LangChain, LangGraph, Text2SQL, RAG, Hugging Face transformers (BERT)
- **Data & Cloud**: Spark, Airflow, Kafka, Trino, Databricks, AWS (S3, Glue), GCP (BigQuery), Azure
- **Databases & Tools**: PostgreSQL, MongoDB, Neo4j, Docker, Git, Linux
- **Spoken Languages**: English (C1), Italian (A2)

Achievements & Test Scores

- **Test Scores**: IELTS 8.0/9.0; GRE Quantitative 165/170
- **Competitive Programming**: ICPC Asia Topi Regionals 2022: 19th place; Google Kickstart (May 2022): 6th place country-wide

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